AIAC Assignmnet-14.4

Name:G.Harika

HT.No:2303A51612

Batch:05

Task 1: Responsive Homepage Layout for a Startup Website

Scenario:

You are designing a basic homepage for a startup company that needs to look good on desktops, tablets, and mobile devices.

Task Description

• Create a web page structure using HTML that includes:

o A header with navigation links

o A main content section

o A footer with basic information

• Use AI assistance to generate responsive CSS using media queries.

• Ensure the layout adjusts smoothly across different screen sizes.

Expected Output

• A responsive web page where:

o Header, content, and footer are clearly separated

o Layout adapts correctly to mobile, tablet, and desktop views

o Code is clean and well-structured

Prompt:

Generate a responsive startup homepage using HTML and CSS.

Requirements:

- Include a header with navigation links (Home, About, Services, Contact)

- A main content section with welcome text

- A footer with copyright information

- Use clean HTML structure

- Use CSS media queries to make the layout responsive for:

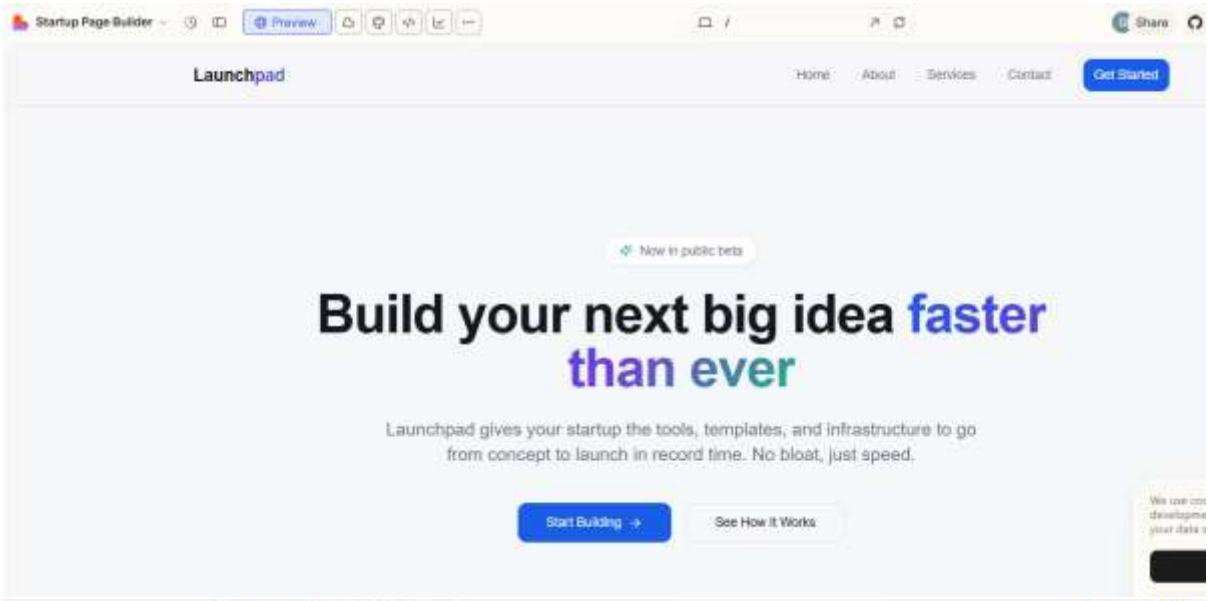
  - Mobile screens

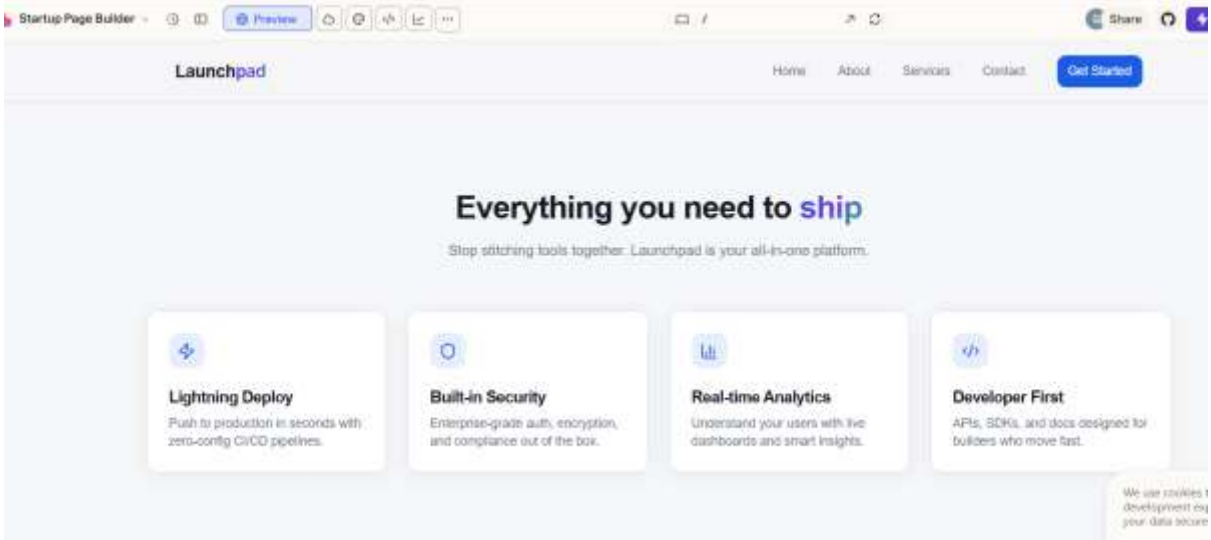
  - Tablets

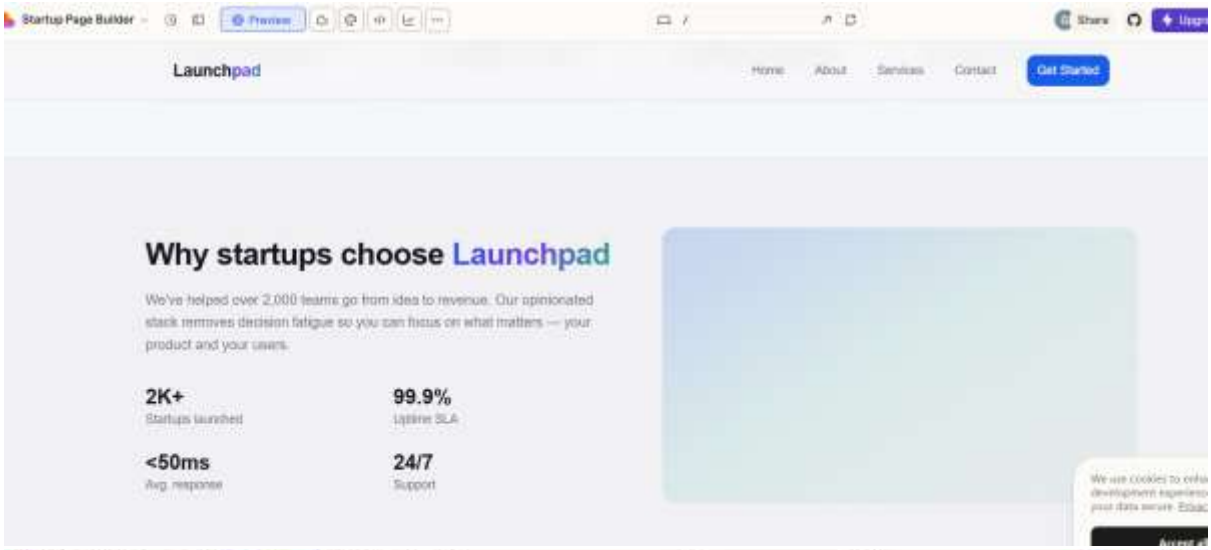
  - Desktop screens

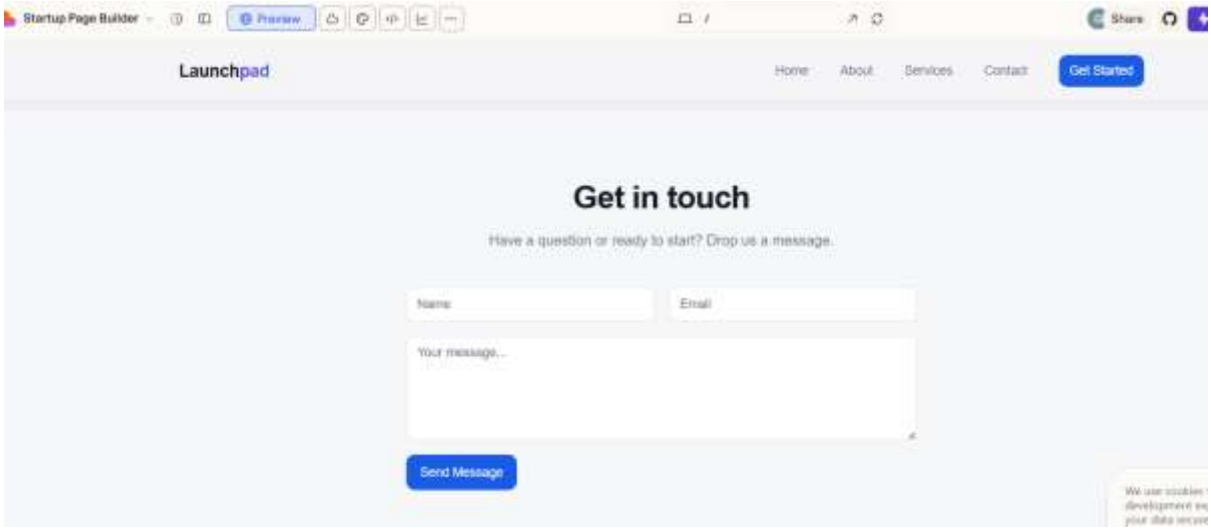
- Ensure header, content, and footer are clearly separated

Output:









Explaination:

This task creates a responsive startup homepage using HTML and CSS.

- HTML is used to create the structure of the webpage with three sections: Header, Main Content, and Footer.
- The Header contains the website title and navigation links like Home, About, Services, and Contact.
- The Main section displays the welcome content of the startup.
- The Footer shows basic information such as copyright.

CSS is used to style the webpage and make it responsive using media queries.

- On desktop screens, the navigation links appear in a horizontal row.
- On tablet screens, the links adjust to fit the screen.
- On mobile screens, the layout becomes vertical and text size reduces for better viewing.

This ensures the webpage looks good and works properly on desktop, tablet, and mobile devices.

Task 2: Interactive Call-to-Action Button

Scenario

A marketing page requires a call-to-action button that responds when users interact with it.

Task Description

• Add a button to a web page using HTML.

• Use AI to generate JavaScript that responds to the button click by:

o Displaying a message or alert

• Ask AI to add meaningful comments explaining the JavaScript logic.

Expected Output:

• A web page where:

o Clicking the button triggers a visible response (alert or message)

o JavaScript code is readable and commented

o No page reload occurs

Prompt:

Create a simple web page with a Call-to-Action button.

Requirements:

- Use HTML to add a button labeled "Get Started"

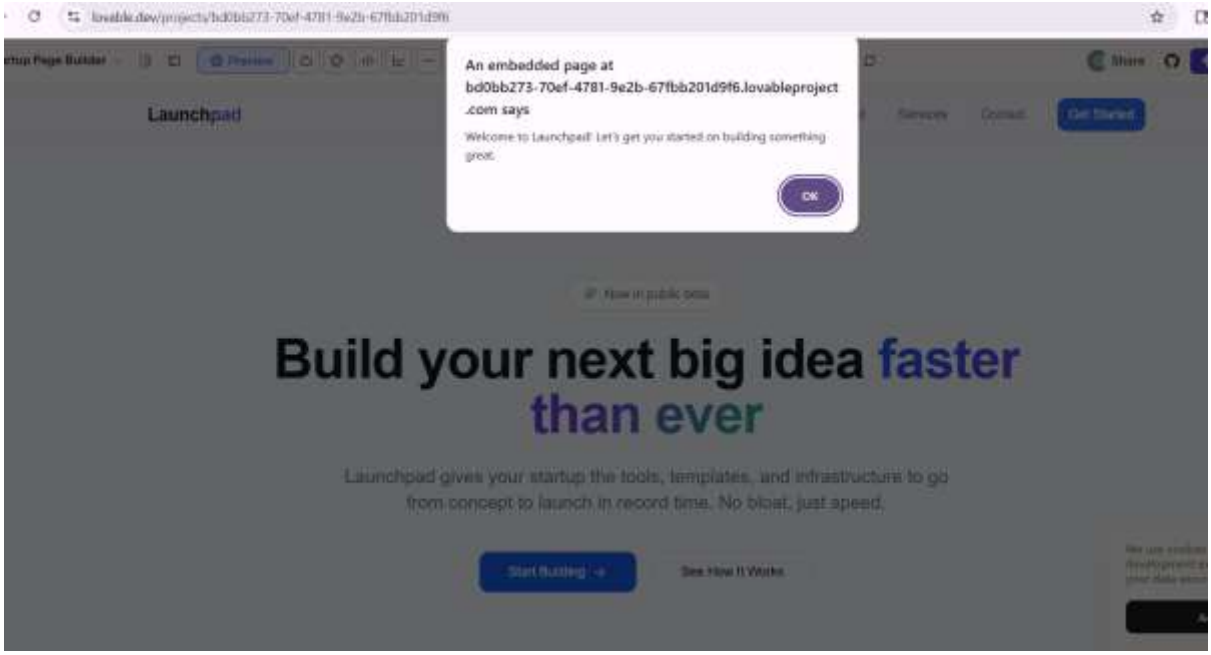
- Use JavaScript to show an alert message when the button is clicked

- Do not reload the page

- Add meaningful comments in the JavaScript explaining the logic

- Keep the code clean and easy to understand

Output:



Explanation:

In this task, a Call-to-Action (CTA) button is added to a web page using HTML. The button encourages users to interact with the page. JavaScript is used to make the button interactive. An event listener is added to the button so that when the user clicks it, a function runs. When the button is clicked, JavaScript displays an alert message such as “Thank you for your interest”. This gives immediate feedback to the user. The JavaScript code also contains comments to explain each step, making the code easy to understand and readable. The interaction happens on the same page, so no page reload occurs.

Task 3: Contact Form with Client-Side Validation

Scenario

You are developing a contact form for a company website to collect

user queries safely and correctly.

Task Description

• Design a form with fields for:

o Name

o Email

o Message

• Use AI to generate JavaScript validation logic that:

o Prevents empty submissions

o Checks for a valid email format

• Display clear error messages for invalid input.

Expected Output:

• A functional form where:

o Invalid input shows inline error messages

o Valid input displays a success message

o Validation happens on the client side using JavaScript

Prompt:

Create a contact form using HTML with the following fields:

- Name

- Email

- Message

Add JavaScript client-side validation that:

- Prevents the form from submitting if fields are empty

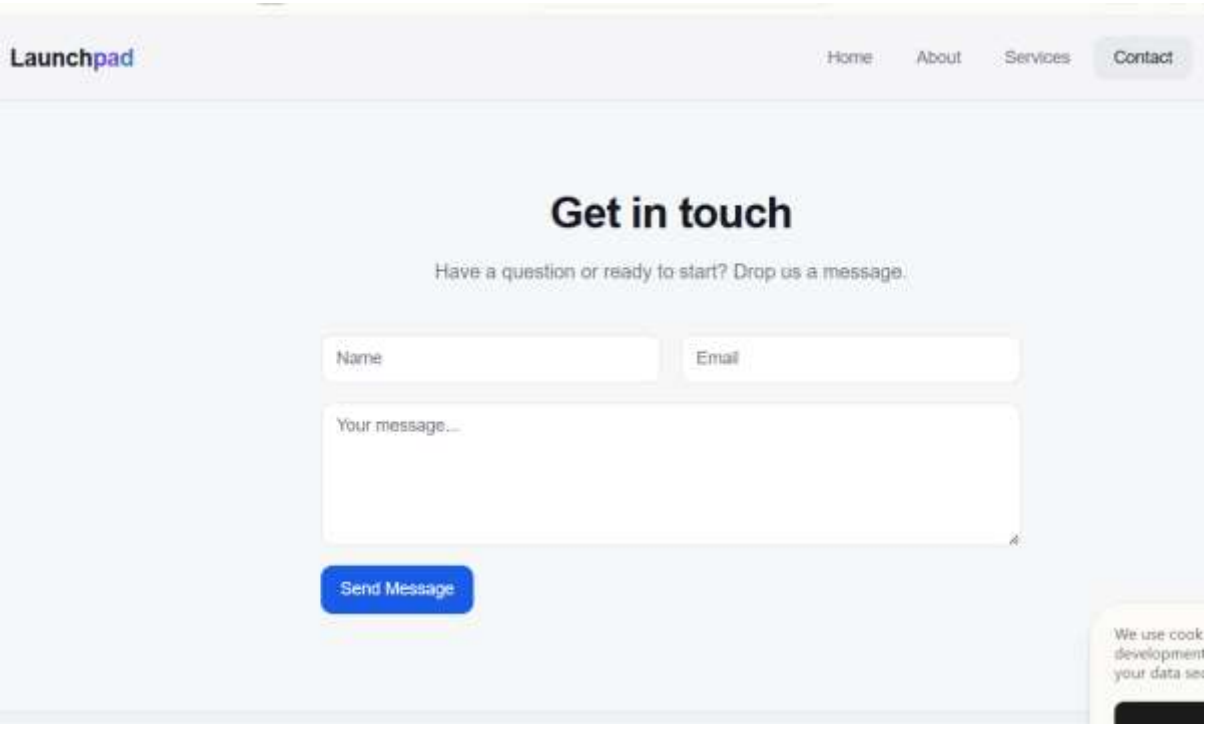
- Checks if the email format is valid

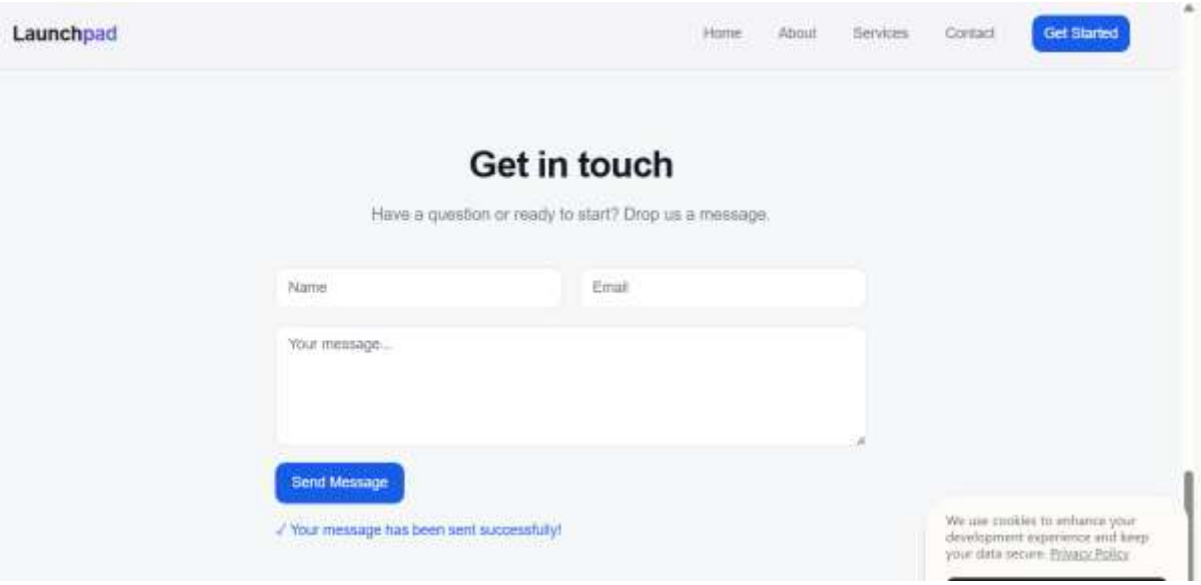
- Displays clear inline error messages below each field

- Shows a success message when all inputs are valid

Add comments in the JavaScript code explaining the validation logic.

Output:





Explanation:

In this task, a contact form is created using HTML with three fields: Name, Email, and Message. This form allows users to send queries to the company.

JavaScript is used for client-side validation to check the user input before the form is submitted.

- If any field is empty, an error message is displayed below that field.
- The email field is checked to ensure it follows a valid email format (example: user@email.com).
- If the email format is incorrect, an error message is shown.

The function prevents form submission until all fields are correctly filled. When the user enters valid information, a success message appears on the page. This validation happens on the client side using JavaScript, so the page does not need to reload or send incorrect data.

Task 4: Dynamic List Management for a Product Page

Scenario

A product listing page needs to add or remove items dynamically

without refreshing the page.

Task Description

• Create a simple HTML list to display items.

• Use AI to generate JavaScript that:

o Adds a new item when an “Add” button is clicked

o Removes the last item when a “Remove” button is clicked

• Ensure the DOM updates dynamically.

Expected Output:

• A web page where:

o Items can be added and removed dynamically

o Changes appear instantly without page reload

o JavaScript logic is clear and maintainable

Prompt:

Create a simple web page that manages a product list.

Requirements:

- Use HTML to create a list of items

- Add two buttons: "Add Item" and "Remove Item"

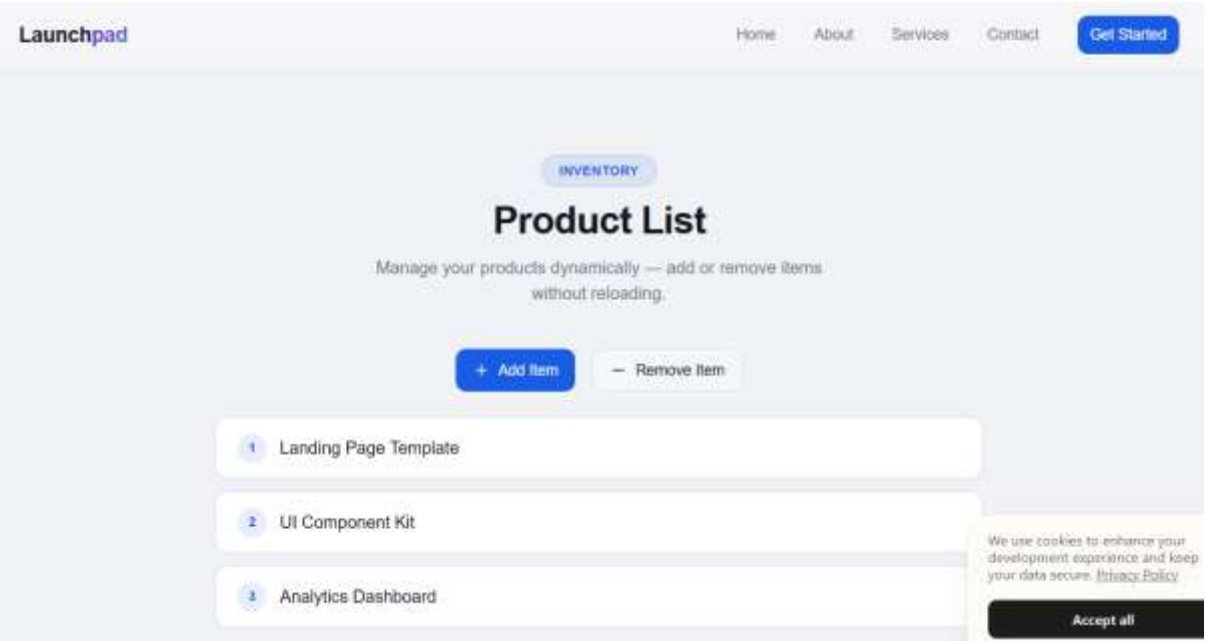
- Use JavaScript to dynamically add a new item to the list when the Add button is clicked

- Remove the last item when the Remove button is clicked

- Ensure the list updates without reloading the page

- Write clear and maintainable JavaScript with comments explaining the logic

Output:



Explanation:

In this task, a product list is created using HTML with items displayed inside an unordered list (<ul>). Two buttons are added: Add Item and Remove Item. JavaScript is used to dynamically manage the list.

- When the Add Item button is clicked, JavaScript creates a new list item (<li>) and adds it to the product list.
- When the Remove Item button is clicked, JavaScript removes the last item from the list.

The changes happen by updating the DOM (Document Object Model) directly, so the list updates instantly without refreshing the page. This makes the product page interactive and dynamic, allowing users to add or remove items easily.

Task 5: Modal Popup for User Notifications

Scenario

You are building a user notification popup for announcements or alerts on a website.

Task Description

• Use AI to help generate:

o HTML structure for a modal popup

o CSS styling with a semi-transparent background overlay

o JavaScript to open and close the modal

• Include multiple ways to close the modal (button or overlay click).

Expected Output:

• A web page where:

o Clicking “Open Modal” displays the popup

o Popup overlays the page content clearly

o Modal closes smoothly when instructed

Prompt:

Create a web page with a modal popup notification.

Requirements:

- Use HTML to create a button called "Open Modal"

- Create a modal popup structure for displaying a notification

- Style the modal using CSS with a semi-transparent background overlay

- Use JavaScript to open and close the modal

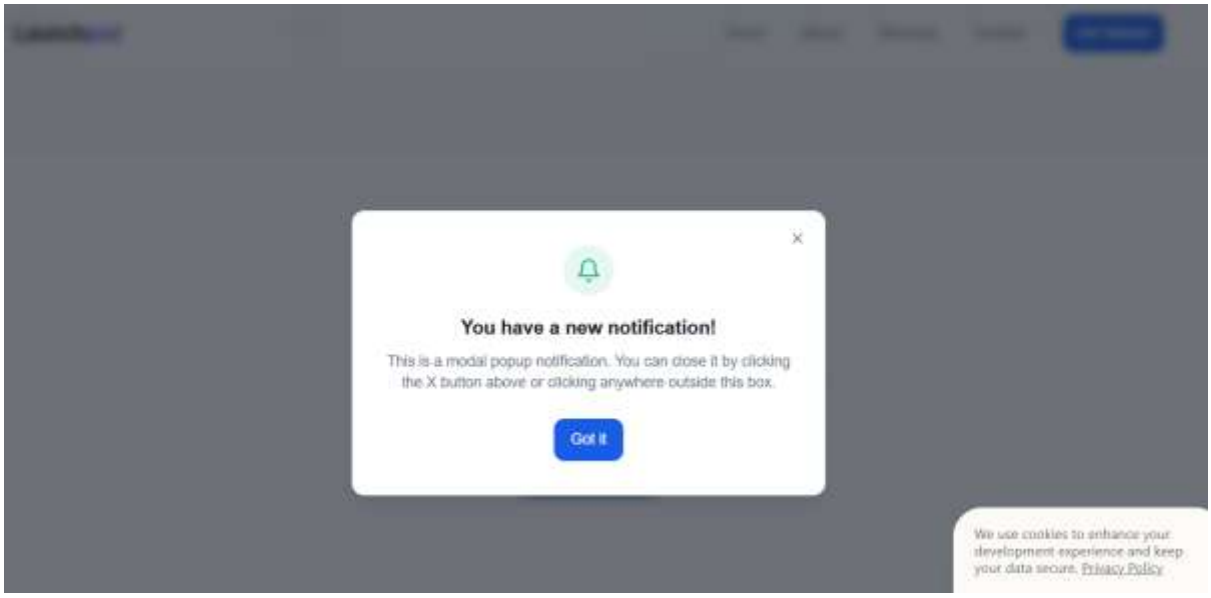
- Allow the modal to close in multiple ways:

  • Clicking the close button

  • Clicking outside the modal (overlay)

- Keep the code simple, readable, and well-structured

Output:



Explanation:

In this task, a modal popup is created to display notifications or announcements on a website. HTML is used to create the structure of the modal, including the Open Modal button, the popup content, and a close (×) button. CSS is used to style the modal. A semi-transparent background overlay is added using rgba() so that the popup appears clearly on top of the webpage while the background becomes slightly dark. JavaScript controls the opening and closing of the modal.

- When the Open Modal button is clicked, the popup appears.
- The modal can be closed by clicking the close (×) button.
- It can also close when the user clicks outside the popup on the overlay.

This creates a smooth user notification system where the popup appears over the page content and closes easily without reloading the page.